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Theoretical Physics

Group Theory

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Introduction

1 | Symmetry in Quantum Physics

1.1 | Symmetry Groups in Quantum Physics

Symmetry transformations are operations carried out on the devices which prepare the states and the apparatuses which measure the observables of a quantum system having the property of not changing the results of experiments. For instance, space translations are symmetry transformations since the uniform translation of all preparations devices and measuring apparatuses does not change the results of experiments due to the homogeneity of space. Likewise, space rotations are symmetry transformations since the uniform rotation of all preparations devices and measuring apparatuses does not change the results of experiments due to the isotropy of space. In relativity, Lorentz boosts are symmetry transformations since as the result of experiments are the same in a given inertial frame and any inertial frame moving at uniform speed with respect to the first because the isotropy of spacetime.

In general, symmetry transformations gather together in families with special properties. Let G be one such family. Carrying out successively two symmetry transformations g, g' of G yields a third symmetry transformation $g' \cdot g$ of G called the product of g, g' . The effect of a symmetry transformation g of G can be undone by another symmetry transformation g^{-1} of G called the inverse of g . There exists a distinguished symmetry transformation e of G that has no effect called neutral. The multiplication and inversion operations and neutral element of G have the following properties.

[...]

Example (Space Translation Symmetry).

Example (Space Rotation Symmetry).

1.1.1 | Symmetry Groups and Their Operator Realization

In quantum theory, states and observables of a system are represented respectively by normalized vectors defined up to a multiplicative phase in a Hilbert space $\mathcal{H}$ and selfadjoint operators in the operator algebra $L(\mathcal{H})$. The invariance of experimental results under the action of a symmetry group reduces to the invariance of expectation values: if Ψ is a state, O is an observable and g is a symmetry transformation of G , ${}^g\Psi$ is the transformed state of Ψ and gO is the transformed observable of O , one must have

$$\langle {}^g\Psi | {}^gO | {}^g\Psi \rangle = \langle \Psi | O | \Psi \rangle .$$

As is well-known, there is a class of elementary observables, the “yes-no” ones, defined by taking the value 1, 0 according to whether the system is in a certain state Φ or not. The associated operator is given by

$$O_{\Phi}\Psi = \Phi \langle \Phi | \Psi \rangle .$$

1.2 | Dynamical Symmetry Groups

1.2.1 | Energy Eigenvalues Classification

1.3 | Lie Groups and Lie Algebras

A Lie group G is an infinite group that can be parametrized in a neighborhood of the neutral element by continuous parameters θ^a , $a = 1, \dots, d$, organized as a vector $\boldsymbol{\theta}$, in one-to-one fashion:

$$\begin{cases} \boldsymbol{\theta} \in D \subseteq \mathbb{R}^d, \\ g : D \mapsto G, \\ \boldsymbol{\theta} \mapsto g(\boldsymbol{\theta}), \end{cases}$$

where D is an open neighborhood of the origin in $\mathbb{R}^d$ ($\mathbf{0} \in D$ and $g(\mathbf{0}) \in G$). The number d of parameters is called the dimension of G , seen as a differential map. The group operations of multiplication and inversion are smooth functions of the parameters.

If the parametrization is smooth, as we assume, G can be thought geometrically as a group manifold. Below, we treat G as a matrix group for definiteness.

We assume that 0 belongs to the parameter space and that

$$g(\mathbf{0}) = 1,$$

a convenient convention. By virtue of the parametrization, for any pair of parameter vectors $\boldsymbol{\theta}$, $\boldsymbol{\theta}'$, there exists a third parameter vector $\boldsymbol{\theta}''$ such that

$$g(\boldsymbol{\theta}')g(\boldsymbol{\theta}) = g(\boldsymbol{\theta}'').$$

As the parametrization is one-to-one,¹ $\boldsymbol{\theta}''$ is also uniquely determined by $\boldsymbol{\theta}$, $\boldsymbol{\theta}'$. So, it can be expressed as a function of them

$$\boldsymbol{\theta}'' = m(\boldsymbol{\theta}', \boldsymbol{\theta}),$$

which is a function of the **group multiplication**. The function m is smooth as the group multiplication is smooth. The inverse of $g(\boldsymbol{\theta})$ is given by

$$g(\boldsymbol{\theta})^{-1} = g(\boldsymbol{\theta}'),$$

for some $\boldsymbol{\theta}' = j(\boldsymbol{\theta})$, which is unique since the parameterization is one-to-one. The function j is smooth as the group inversion is smooth.

[...]

These functions have to respect some properties, which are the group multiplication properties: associativity, invertibility and neutrality. Thus given three parameter vectors $\boldsymbol{\theta}$, $\boldsymbol{\theta}'$, $\boldsymbol{\theta}''$, the following relations must hold:

$$\begin{aligned} (g(\boldsymbol{\theta})g(\boldsymbol{\theta}'))g(\boldsymbol{\theta}'') &= g(\boldsymbol{\theta})(g(\boldsymbol{\theta}')g(\boldsymbol{\theta}'')), \\ g(m(\boldsymbol{\theta}, \boldsymbol{\theta}'))g(\boldsymbol{\theta}'') &= g(\boldsymbol{\theta})g(m(\boldsymbol{\theta}', \boldsymbol{\theta}'')), \\ g(m(m(\boldsymbol{\theta}, \boldsymbol{\theta}'), \boldsymbol{\theta}'')) &= g(m(\boldsymbol{\theta}, m(\boldsymbol{\theta}', \boldsymbol{\theta}''))), \end{aligned}$$

¹This means that each element of the group corresponds to a unique set of parameters: $g(\boldsymbol{\theta}) = g(\boldsymbol{\theta}') \implies \boldsymbol{\theta} = \boldsymbol{\theta}'$.

since the parametrization is one-to-one, these relations [...] and so we find the three properties of the group multiplication functions:

$$\begin{aligned} m(m(\boldsymbol{\theta}, \boldsymbol{\theta}'), \boldsymbol{\theta}'') &= m(\boldsymbol{\theta}, m(\boldsymbol{\theta}', \boldsymbol{\theta}'')), \\ m(\boldsymbol{\theta}, j(\boldsymbol{\theta})) &= m(j(\boldsymbol{\theta}), \boldsymbol{\theta}) = \mathbf{0}, \\ m(\boldsymbol{\theta}, \mathbf{0}) &= m(\mathbf{0}, \boldsymbol{\theta}) = \boldsymbol{\theta}. \end{aligned} \tag{1.3.1}$$

The idea now is to try and understand the structure of G by looking at the behavior of $g(\boldsymbol{\theta})$ in a neighborhood of the neutral element. This is done by Taylor expanding it in power series around $\mathbf{0}$:

$$g(\boldsymbol{\theta}) = g(\mathbf{0}) + \left. \frac{\partial g}{\partial \theta^a} \right|_{\mathbf{0}} \theta^a + \frac{1}{2} \left. \frac{\partial^2 g}{\partial \theta^a \partial \theta^b} \right|_{\mathbf{0}} \theta^a \theta^b + O(\theta^3).$$

Now computing individual terms of the expansion, we have

$$g(\boldsymbol{\theta}) = 1, \quad t_a = \left. \frac{\partial g}{\partial \theta^a} \right|_{\mathbf{0}}, \quad t_{ab} = \left. \frac{\partial^2 g}{\partial \theta^a \partial \theta^b} \right|_{\mathbf{0}},$$

so that

$$g(\boldsymbol{\theta}) = 1 + t_a \theta^a + \frac{1}{2} t_{ab} \theta^a \theta^b + O(\theta^3).$$

Let us highlight the fact that the coefficients t_a, t_{ab} are matrices, as $g(\boldsymbol{\theta})$ is a matrix group. t_{ab} in particular is symmetric in its indices a, b since the order of differentiation does not matter.

We can now think of G as an **hypersurface in the space of matrices**, containing the identity matrix. In the neighborhood of the identity, there is D , subset of our hypersurface, which is diffeomorphic to an open neighborhood of the origin in $\mathbb{R}^d$.

If we vary $g(0, \dots, 0, \theta^a, 0, \dots, 0)$, we get a curve in D passing through the identity (when $\theta^a = 0$). The tangent vector to this curve at the identity is given by $t_a = \left. \frac{\partial g}{\partial \theta^a} \right|_{\mathbf{0}}$. Varying all parameters θ^a we get d curves in D passing through the identity and their tangent vectors at the identity are given by $t_a, a = 1, \dots, d$. These tangent vectors are **linearly independent** since the parametrization is one-to-one. Thus, they span a vector space of dimension d .

Thus the tangent space to G at the identity is a vector space of dimension d spanned by the matrices $t_a, a = 1, \dots, d$, constituting a basis. We denote this tangent space by $\mathfrak{g}$ and call it the Lie algebra of G . It is a vector space of dimension d and it is closed under the commutator of matrices. The Lie algebra $\mathfrak{g}$ encodes all the information about the local structure of G around the identity element.

We could now think to Taylor expand the group multiplication functions m, j around $\mathbf{0}$ and try to find the structure of G by looking at the properties of the coefficients of the expansion. Starting from the associativity property, we find that the coefficients of the expansion of m must satisfy

$$m^a(\boldsymbol{\theta}, \boldsymbol{\theta}') = \theta^a + \theta'^a + m_{bc}^a \theta^b \theta'^c + O(\theta'^2, \boldsymbol{\theta}) + O(\boldsymbol{\theta}^2, \boldsymbol{\theta}'),$$

while for the inversion property we find

$$j^a(\boldsymbol{\theta}) = -\theta^a + j_{bc}^a \theta^b \theta^c + O(\theta^3).$$

If we now recall the Taylor expansion of $g(\boldsymbol{\theta})$ around $\mathbf{0}$

$$g(\mathbf{0}) = 1 + t_a \theta^a + \frac{1}{2} t_{ab} \theta^a \theta^b + O(\theta^3),$$

we can note how the coefficients t_a define another member of the Lie algebra $\mathfrak{g}$ by the commutator

$$[t_a, t_b] = t_a t_b - t_b t_a = f_{ab}^c t_c \in \mathfrak{g},$$

where f_{ab}^c are the **structure constants** of the Lie algebra $\mathfrak{g}$. The structure constants are *antisymmetric in their lower indices* a, b since the commutator is antisymmetric. These structure constants are related to the coefficients m_{bc}^a of the expansion of the group multiplication function m by

$$f_{bc}^a = m_{bc}^a - m_{cb}^a.$$

If we change parametrization, the structure constants change accordingly, but the Lie algebra $\mathfrak{g}$ remains the same.

Taking again the definition of the multiplication function

$$g(\theta')g(\theta) = g(m(\theta', \theta)),$$

we can Taylor expand both sides around $\mathbf{0}$ and find the following relation between the coefficients of the expansion of g and the structure constants:

$$\begin{aligned} \left[1 + \theta^a t_a + \frac{1}{2} \theta^a \theta^b t_{ab} + O(\theta^3) \right] \left[1 + \theta'^a t_a + \frac{1}{2} \theta'^a \theta'^b t_{ab} + O(\theta'^3) \right] \\ = [\dots] \end{aligned}$$

and by relating the coefficients of the expansion of both sides in θ^b, θ'^a we find the following relations (renaming summation indices):

$$\begin{aligned} t_a t_b &= t_{ab} + m_{ab}^c t_c, \\ t_b t_a &= t_{ab} + m_{ba}^c t_c, \end{aligned}$$

and if we subtract the second from the first, we find

$$[t_a, t_b] = (m_{ab}^c - m_{ba}^c) t_c = f_{ab}^c t_c,$$

so that the structure constants f_{ab}^c of the Lie algebra $\mathfrak{g}$ are related to the coefficients m_{ab}^c of the expansion of the group multiplication function m by

$$f_{ab}^c = m_{ab}^c - m_{ba}^c.$$

Thus to sum up:

- The Lie algebra $\mathfrak{g}$ of a Lie group G is the tangent space to G at the identity element, spanned by the matrices t_a , $a = 1, \dots, d$.
- The basis elements t_a of the Lie algebra $\mathfrak{g}$ satisfy the commutation relations

$$[t_a, t_b] = f_{ab}^c t_c,$$

where f_{ab}^c are the structure constants of the Lie algebra $\mathfrak{g}$.

- The structure constants f_{ab}^c of the Lie algebra $\mathfrak{g}$ are related to the coefficients m_{ab}^c of the expansion of the group multiplication function m by

$$f_{ab}^c = m_{ab}^c - m_{ba}^c.$$

The Lie algebra, the tangent space to the Lie group at the identity, is not only a vector space, but also an algebra, since it is closed under the commutator of matrices. The Lie algebra encodes all the information about the local structure of the Lie group around the identity element. In particular, it encodes the structure constants f_{ab}^c which determine the commutation relations of the basis elements t_a of the Lie algebra.

The commutator of the basis elements t_a of the Lie algebra $\mathfrak{g}$ has some properties, which are the **Jacobi identity** and the **antisymmetry** of the commutator:

$$\begin{aligned} [t_a, t_b] + [t_b, t_a] &= 0, \\ [t_a, [t_b, t_c]] + [t_b, [t_c, t_a]] + [t_c, [t_a, t_b]] &= 0. \end{aligned}$$

These two properties can be expressed in terms of the structure constants f_{ab}^c as follows:

$$\begin{aligned} f_{ab}^c + f_{ba}^c &= 0, \\ f_{ad}^e f_{bc}^d + f_{bd}^e f_{ca}^d + f_{cd}^e f_{ab}^d &= 0, \end{aligned}$$

which are the **antisymmetry** of the structure constants in their lower indices and the **Jacobi identity** for the structure constants, respectively. This means that not every set of numbers f_{ab}^c can be the structure constants of a Lie algebra, but only those that satisfy the antisymmetry and Jacobi identity properties.

1.3.1 | Exponential Parametrization of Lie Groups

Given the parametrization of a Lie group G in a neighborhood of the identity element we used to define the Lie algebra $\mathfrak{g}$ of G

$$g : D \mapsto G, \quad \theta \in D \mapsto g(\theta) \in G,$$

we can now try to find a global parametrization of G in terms of the elements of $\mathfrak{g}$. If we consider another parametrization of G in a neighborhood of the identity element defined by

$$\tilde{g}(\theta) = \lim_{N \rightarrow \infty} g\left(\frac{\theta}{N}\right)^N,$$

we obtain the **exponential map**, which provides a global parametrization of the Lie group G in terms of the elements of its Lie algebra $\mathfrak{g}$. Let us analyze its behaviour in a neighborhood of the identity element. We have

$$g\left(\frac{\theta}{N}\right)^N = \left[1 + \frac{t_a \theta^a}{N} + \frac{1}{2} \left(\frac{t_a \theta^a}{N}\right)^2 + O\left(\frac{1}{N^3}\right)\right]^N \sim \left(1 + \frac{t_a \theta^a}{N}\right)^N \xrightarrow{N \rightarrow \infty} e^{t_a \theta^a},$$

which is the exponential of a matrix, defined by the power series

$$e^{t_a \theta^a} = \sum_{n=0}^{\infty} \frac{(t_a \theta^a)^n}{n!}.$$

Thus, the exponential map provides a global parametrization of the Lie group G in terms of the elements of its Lie algebra $\mathfrak{g}$ as follows:

$$\tilde{g}(\theta) = e^{t_a \theta^a} = 1 + \tilde{t}_a \theta^a + \frac{1}{2} \theta^a \theta^b \tilde{t}_{ab} + O(\theta^3),$$

where

$$\tilde{t}_A = t_a, \quad \tilde{t}_{ab} = t_a t_b + t_b t_a.$$

The same goes for the group multiplication function m and the inversion function j . We can define new functions $\tilde{m}, \tilde{j}$ as follows:

$$\begin{aligned}\tilde{j}(\theta) &= -\theta^a, \\ \tilde{g}(\tilde{j}(\theta)) &= \tilde{g}(\theta)^{-1} = \dots = e^{-t_a \theta^a},\end{aligned}$$

while for the multiplication function we have the **Campbell-Baker-Hausdorff function** so that

$$[\dots]$$

These new structure constants are not different from the original ones, since the exponential map is a reparametrization of the Lie group G in terms of the elements of its Lie algebra $\mathfrak{g}$: $\tilde{t}_a = t_a$ implies that

$$f_{ab}^c t_c = [t_a, t_b] = [\tilde{t}_a, \tilde{t}_b] = \tilde{f}_{ab}^c \tilde{t}_c = \tilde{f}_{ab}^c t_c,$$

thus the structure constants of the Lie algebra $\mathfrak{g}$ are the same in both parametrizations, as expected since the Lie algebra is an intrinsic property of the Lie group G and does not depend on the choice of parametrization:

$$f_{ab}^c = \tilde{f}_{ab}^c.$$

Thus we have

$$\tilde{m}_{bc}^a = \frac{1}{2} \tilde{f}_{bc}^a = \frac{1}{2} f_{bc}^a,$$

and if we expand the CBH function in power series we find²

$$\tilde{m}^a(\theta, \theta') = \theta^a + \theta'^a + \frac{1}{2} f_{bc}^a \theta^b \theta'^c + O(\theta^3, \theta'^3),$$

which is like a polynomial expansion of the group multiplication function m in terms of the structure constants f_{bc}^a of the Lie algebra $\mathfrak{g}$. The exponential map thus provides a global parametrization of the Lie group G in terms of the elements of its Lie algebra $\mathfrak{g}$ and allows us to express the group multiplication function m in terms of the structure constants f_{bc}^a of the Lie algebra $\mathfrak{g}$.

Example (Phase Group U(1)). Let us define the group of phase transformations U(1) as the group of complex numbers of unit modulus under multiplication:

$$U(1) = \{z \in \mathbb{C} : |z| = 1\} = \{e^{i\theta} : \theta \in \mathbb{R}\}.$$

It is an abelian Lie group of dimension 1 which represents the Gauge group of QED. The elements of U(1) can be represented by an angle θ with an exponential parametrization as follows:

$$\tilde{g}(\theta) = e^{i\theta^a t_a} = e^{i\theta},$$

since the Lie algebra of U(1) is one-dimensional and spanned by the identity matrix $t = 1$. The structure constants of the Lie algebra of U(1) are zero since the commutator of any two elements of the Lie algebra is zero:

$$[t, t] = 0 \implies f_{ab}^c = 0.$$

²There exists a closed form expression for the CBH function, but it is not important for our purposes: remind only that exact complicated expressions for computing all the terms exist, but we are only interested in the first few terms of the expansion.

Example (Special Unitary Group $SU(2)$). The special unitary group $SU(2)$ is the group of 2×2 unitary matrices with determinant 1:

$$SU(2) = \{U \in \mathbb{C}^{2 \times 2} : U^\dagger U = \mathbb{I}_2, \det(U) = 1\}.$$

It is a non-abelian Lie group of dimension 3: $\theta \in \mathbb{R}^3$. The elements of $SU(2)$ can be parametrized using the Pauli matrices σ_i as follows:

$$\tilde{g}(\theta) = e^{i\theta^i \frac{\tau(\theta)}{2}},$$

where

$$\tau(\theta) = \begin{pmatrix} \theta^3 & \theta^1 - i\theta^2 \\ \theta^1 + i\theta^2 & -\theta^3 \end{pmatrix} = \theta^1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + \theta^2 \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + \theta^3 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \theta^i \sigma_i,$$

Thus $t_a = \frac{1}{2}\tau_a$ and the structure constants of the Lie algebra $\mathfrak{su}(2)$ are given by the Levi-Civita symbol ϵ_{ijk} :

$$[\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k \implies f_{ij}{}^k = 2\epsilon_{ijk}.$$

Example (Special Orthogonal Group $SO(3)$). The special orthogonal group $SO(3)$ is the group of 3×3 real orthogonal matrices with determinant 1:

$$SO(3) = \{R \in \mathbb{R}^{3 \times 3} : R^T R = \mathbb{I}_3, \det(R) = 1\}.$$

It is a non-abelian Lie group of dimension 3. The elements of $SO(3)$ can be parametrized using the generators of rotations J_i as follows:

$$\tilde{g}(\theta) = e^{i\theta^i J_i},$$

where the generators J_i satisfy the commutation relations

$$[J_i, J_j] = i\epsilon_{ijk}J_k.$$

[....-] to be finished

Appendices

A | Notation and Conventions

B | Computations and further developments